No second subgenome more divergent than diploid alleles in the deposited genome of a tetraploid stinging nettle (*Urtica dioica*): a preregistered genome-wide test against haplotype-resolved diploid assemblies

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Abstract

Stinging nettle (*Urtica dioica*) has a diploid (2n = 26) and a dominant tetraploid (2n = 52) cytotype, and whether the tetraploid is an autopolyploid or an allopolyploid has not been tested genome-wide. Using public chromosome-level assemblies — haplotype-resolved genomes of a Romanian diploid and of her diploid son by a Bulgarian father, and the Darwin Tree of Life assembly of a UK tetraploid — we preregistered a test before downloading any sequence. Across 13,432 single-copy genes, the median divergence of the tetraploid primary pseudo-haplotype from the nearest Romanian haplotype was 1.11 times that of the unrelated Bulgarian haplotype (95 % CI 1.06–1.16), and 1.6 % of genes exceeded the diploid 99th percentile, against ≥ 20 % under the preregistered allotetraploid criterion; the held-out alternate pseudo-haplotype gave 1.10. The few divergent genes cluster where the two diploids share a near-identical haplotype, not where the tetraploid is unusual; the two tetraploid pseudo-haplotypes differ less than the diploid's own alleles, and retained second gene copies are at allelic identity. These data exclude a second subgenome diverged by ≥ 0.5 % and contributing ≥ 25 % of deposited genes, or by ≥ 2 % and contributing ≥ 10 %, against 9.4 % divergence from *U. urens*. Within the deposited sequence the tetraploid is therefore drawn from the Eurasian diploid gene pool; because that sequence holds 68 % of the tetraploid's read k-mers, a purged divergent subgenome cannot be excluded without the read-based analysis we specify. An independent nanopore assembly described as diploid has gene copy numbers intermediate between diploid and tetraploid expectations; its ploidy is unresolved. Two preregistered secondary hypotheses, on centromere-type-specific diversity and on the polarity of the chromosome 8/13 rearrangement, were respectively contradicted and inconclusive.

Introduction

Urtica dioica is a mixed-ploidy species complex. Flow-cytometric screening of 7,012 plants from 1,317 European and south-west Asian populations found 87 % tetraploids and 13 % diploids, with chromosome counts of 2n = 26 and 2n = 52 for the two cytotypes and a uniform monoploid genome size (Cx ≈ 0.55–0.59 pg) across the complex (Rejlová et al., 2019). Diploids are the named Eurasian subspecies (*subinermis*, *sondenii*, *pubescens*, *kurdistanica*) and occupy natural and near-natural habitats; the tetraploid subsp. *dioica* is the synanthropic, nitrophilous nettle of disturbed ground across the Holarctic (Rejlová et al., 2019; Urfus et al., 2021; Grosse-Veldmann & Weigend, 2015; Taylor, 2009). The tetraploid "is also often supposed to be an allopolyploid or a group of allopolyploids with different evolutionary histories" (Rejlová et al., 2019), a view the same authors trace to Geltman (1990, Botanicheskii Zhurnal 75:840–845), while also asking whether "(auto)polyploidization has taken place" within subsp. *subinermis*. The strongest molecular evidence to date is a Hyb-Seq study of 279 plants (Rejlová et al., 2021): in a coalescent species tree from 249 nuclear genes, the European tetraploids and all four diploid subspecies collapsed into a single polytomy, 2,135 independent SNPs showed no structure among diploids, and "tetraploids coalesced with diploids in both morphological and molecular analyses". Those authors judged the tetraploids co-occurring with diploid subsp. *pubescens* in the Po basin "probably of autopolyploid origin", but concluded that "the evolutionary history of the remaining European tetraploids remains unresolved", with participation of other diploid subspecies "in either the auto- or allopolyploid origin of tetraploids" left open. Their design used consensus sequences per

individual, so alleles and homoeologs were pooled and a divergent subgenome would have been averaged away rather than detected. No genome-wide, assembly-based measurement of subgenome divergence has been published.

Three recent assemblies make such a measurement possible. Hirabayashi et al. (2025) produced a haplotype-resolved, chromosome-level assembly of a diploid female (clone 11-4, Romania; haplotypes FH1 and FH2, 575 and 521 Mb, 13 chromosomes each, heterozygosity ≈ 1.5 %). The same group assembled her diploid son (clone 48M, father 7-5 from Bulgaria; haplotypes MH1 and MH2) and, exploiting the pedigree, assigned each male chromosome to maternal or paternal origin (Hirabayashi et al., 2026); MH2 carries the NCBI RefSeq annotation RS_2026_09 (23,586 protein-coding genes). Independently, the DToL project assembled a tetraploid male from Kew with HiCanu, purge_dups and YaHS; its 13 pseudomolecules represent "one subgenome of the tetraploid genome", curation proceeded "initially on all four haplotypes and subsequently on a pseudo-primary haplotype 1", and the primary and alternate pseudo-haplotypes together contain 68 % of the tetraploid's reliable read k-mers (primary 55 %, alternate 49 %), with 15 % duplicated BUSCOs that the note's reviewers attribute to tetraploidy (Christenhusz et al., 2025). Together these provide three independent diploid haplotypes from two populations (FH1, FH2 and the paternal haplotype of the male) and two pseudo-haplotypes of a tetraploid.

The logic of the test is simple. If the tetraploid formed within the diploid lineage, each of its haplotypes is a sample from the diploid gene pool, and its divergence from a diploid haplotype should match the divergence between two unrelated diploid haplotypes. If it is an allotetraploid, half of its gene copies descend from a second lineage; a pseudo-haplotype that mixes subgenomes will show a spatially clustered class of genes far more divergent than any diploid–diploid comparison, and a pseudo-haplotype that samples mainly the diploid-like subgenome will still reveal the other subgenome as second loci at intermediate identity or as a divergent mode in the primary-versus-alternate comparison. Because the diploid data themselves provide the reference distribution — within-individual (FH1 vs FH2) and between-population (Bulgarian paternal haplotype vs Romanian female) divergence — the test needs no mutation-rate calibration; the species comparison with *U. urens* (Christenhusz et al., 2024b) anchors the scale. The test is blind, by construction, to sequence that is absent from the deposited assemblies, a limitation we quantify below.

We preregistered this test, together with two secondary hypotheses on the structural biology of the nettle genome, before downloading any sequence data, with explicit decision thresholds, a held-out replicate, a negative control and the artefact checks we would run (preregistration file with SHA-256 hash in the Supplementary Material). The manuscript was then subjected to an independent reproduction and an independent hostile review, and every change made in response is declared in the Methods. The secondary hypotheses concerned (i) whether the within-genome contrast between the five chromosomes predicted by RepeatOBserver (Elphinstone et al., 2025) to be acrocentric and the seven predicted to be polycentric (Hirabayashi et al., 2025, 2026) is visible in the nucleotide-diversity landscape, as expected if polycentric chromosomes lack a single large pericentromeric low-recombination domain (Castellani et al., 2024; Hofstatter et al., 2022; Macas et al., 2023), and (ii) whether the karyotype difference between *U. dioica* (n = 13; chromosomes 8 and 13) and *U. urens* (n = 12; chromosome 2) (Hirabayashi et al., 2025) is a fission in the *U. dioica* lineage or a fusion in the *U. urens* lineage.

Results

A preregistered gene-space divergence test

We used the 23,131 longest RefSeq coding sequences per gene from the male H2 annotation as queries and located each in seven assemblies by discontiguous megablast (Table 1; Methods). A query-anchored alignment was built per gene from the best locus in each assembly, and p-distances were computed on columns where both targets carry an unambiguous base, after trimming 10 bp from internal HSP ends and masking positions at which the self-search did not recover the query base. A gene entered the analysis only if it was single-copy (second-locus bitscore < 0.8 × best) with ≥ 80 % CDS coverage in all of FH1, FH2, MH1, MH2 and the tetraploid target.

Three controls established the assay's behaviour before the decisive comparison was made. First, the maternally inherited male haplotype, a recombinant copy of FH1/FH2, was at p-distance exactly 0 from its nearest female haplotype in 98.8 % of 16,819 genes, so the technical floor is negligible. Second, within-individual divergence between the two female haplotypes (W; median 0.47 %, 99th percentile 3.42 %) and between-population divergence of the Bulgarian paternal haplotype from the nearest female haplotype (b; median 0.45 %, 99th percentile 2.77 %) were nearly identical, consistent with the absence of population structure among Eurasian diploids (Rejlová et al., 2021). Third, *U. urens* genes were at median 9.4 % from *U. dioica*, 20 times the within-species value.

The deposited tetraploid pseudo-haplotypes lie inside the diploid gene pool

For 13,432 genes passing all filters (after removing 6 genes whose tetraploid locus was > 5 percentage points more divergent from *U. urens* than the diploid ortholog, a paralog screen added after review), the tetraploid primary pseudo-haplotype's divergence from the nearest female haplotype (t) had median 0.50 % against median b of 0.45 % for the Bulgarian diploid haplotype measured against the same two reference haplotypes: R = 1.106 (chromosome-level bootstrap 95 % CI 1.061–1.163; 2-Mb-block CI 1.077–1.142). The median paired difference t − b was 0.00 % and the mean 0.087 %. 1.62 % of genes (CI 1.36 %–2.06 %) exceeded the 99th percentile of b (2.69 %), against 1 % expected for identical distributions and ≥ 20 % under the preregistered allotetraploid criterion. The whole distribution of t overlies those of W and b (Figure 1a); its upper tail lies within the diploid's own within-individual tail (W exceeds Q99(b) in 1.5 % of the same genes). On the chromosomes where the design is fully symmetric (3, 5 and 6, where the paternal haplotype is MH1 rather than the query haplotype), R = 1.052 (n = 2,568); elsewhere 1.119. HSP end-trimming did not drive the result: with no trimming R = 1.114 and f99 = 1.74 % on 13,539 genes. By the preregistered decision rule the result is "same-lineage tetraploid" (Figure 1b, c); the preregistered held-out replicate — the alternate pseudo-haplotype, not examined until the primary analysis had been written up — gave R = 1.097 (CI 1.065–1.137) and f99 = 1.79 % (CI 1.45 %–2.20 %) over 13,404 genes.

The genes above Q99(b) are not randomly placed. In both pseudo-haplotypes they form more adjacent pairs along the chromosomes than expected (30 vs 3.9 expected in the primary, p = 0.0005; 46 vs 4.5 in the alternate, p = 0.0005; within-chromosome permutation). We therefore scanned the paired difference t − b in 50-gene windows against the genome-wide maximum under within-chromosome permutation (99th percentile 0.34 %). Two regions exceed it in the primary (Figure 5): chromosome 7 at 0.8–22.8 Mb and chromosome 8 at 10.9–13.9 Mb. In neither is t unusual (medians 0.62 % and 0.58 %); rather, b is deflated (0.16 % and 0.16 %, with 22 % and 20 % of genes at exactly zero) because the Bulgarian paternal haplotype is nearly identical to one Romanian female haplotype over these blocks. On chromosome 7 the three diploid haplotypes and the tetraploid alternate pseudo-haplotype share a low-diversity haplotype (W 0.15 %) while the primary carries a different one at ordinary allelic divergence; on chromosome 8 the block overlaps the inversion-rich region that distinguishes nettle haplotypes (Hirabayashi et al., 2025, 2026), and a shared Romanian–Bulgarian haplotype block with almost no diversity is what suppressed recombination across an inversion polymorphism would produce. Across all 13 chromosomes, no 50-gene window has a median t above 1.25 %, and no window shows t elevated while b and W are normal (Figure 5). A residual class of 80 (primary) and 82 (alternate) genes with t > 5 % (0.6 % of each set) survives the paralog screen; these are hypervariable genes whose diploid alleles are themselves 1–2 % apart (median b 1.0 % and 1.9 %), scattered over all chromosomes, and removing them changes R to 1.100 / 1.093 and f99 to 1.32 % / 1.51 % (results/tables/h1_residual_t_gt5_sensitivity.json). The high-t genes themselves have elevated b (fast-evolving, highly polymorphic genes: their median distance to *U. urens* is 16 % against a genome-wide 9.3 %, in the diploid and the tetraploid alike, and their median b is 1.15 % against 0.45 %), and they are not paralogous hits (none of the 218 failed the paralog screen).

The within-tetraploid comparison closes the remaining assembly-level escape route. Primary and alternate pseudo-haplotypes differed by a median of 0.33 % over 13,246 genes, with 2.9 % of genes above Q99(b); on the log scale the distribution is unimodal (Hartigan dip p = 0.99), as are W (p = 0.98) and b (p = 1.00) — the raw-scale dip test rejects unimodality for all three alike because of the spike at zero in discrete p-distances. An allotetraploid would place homoeologous pairs in a second mode near the subgenome divergence; none exists (Figure 2b). Because the alternate is by construction the deposited sequence most similar to the primary at each locus, this comparison is a lower bound on

within-tetraploid divergence and not a measure of tetraploid heterozygosity.

Second loci, copy structure, and the power of the test

A pseudo-haplotype that mixes subgenomes could hide a divergent subgenome in second loci rather than best loci. In the primary assembly, 2,639 of 16,538 genes (16.0 %) had a second locus at ≥ 97 % identity and ≥ 80 % coverage, against 6–7 such genes in each diploid haplotype assembly — retained homologous or homoeologous copies. These second loci were 98.96 % identical (median; 10th percentile 97.94 %) to the diploid coding sequence, i.e. at allelic divergence, and the best-locus divergence of genes with and without a retained second copy was identical (median t 0.50 % vs 0.50 %). Second loci at 90–97 % identity, where a homoeolog diverged by 3–10 % would appear, were no more frequent in the tetraploid pseudo-haplotypes (10.3 % and 9.9 % of all queries, ≥ 80 % coverage) than in the four diploid haplotype assemblies (9.4–9.8 %), nor were second loci at 80–90 % (4.9–5.2 % vs 5.4–5.6 %); only the allelic class (≥ 97 %) is enriched (22.3 % and 21.4 % vs 9.6–12.2 %), as expected when four haplotypes are squeezed into two pseudo-haplotypes. Across the copy-number spectrum (Figure 2a), the two DToL pseudo-haplotypes together hold ≥ 3 near-identical loci for 32.1 % of genes, against 11.5 % (FH1 + FH2) and 13.3 % (MH1 + MH2) when both haplotypes of a diploid are combined.

What, then, do the data exclude? We simulated a subgenome contributing a fraction φ of the deposited genes with extra divergence δ added to a baseline resampled from b, and compared the expected R and f99 with the observed chromosome-level upper confidence limits (Table 3). Every combination with δ ≥ 0.5 % and φ ≥ 0.25, with δ ≥ 2 % and φ ≥ 0.10, or with δ = 0.25 % and φ ≥ 0.5, is excluded. Not excluded are a subgenome contributing ≤ 10 % of the deposited genes at up to 1.5 % divergence, or ≤ 25 % at 0.25 %. Divergences of 0.25–1.5 % are within the range of present-day diploid alleles (W 99th percentile 3.4 %), so the data are compatible with autotetraploidy, or with a tetraploid formed between diploid populations no more differentiated than the alleles segregating in one Romanian plant, and incompatible with an allotetraploid whose second subgenome is represented in the deposited sequence at ≥ 10 % with ≥ 2 % divergence. On the *U. urens* scale (9.4 % for the species split), 2 % is roughly a fifth of the way to a separate species.

The Bonn nanopore assembly: copy structure intermediate between diploid and tetraploid, ploidy unresolved

While this work was in progress, Wolff et al. (2026) released a 1.1-Gb nanopore assembly of a German nettle (collected in Braunschweig) with a mapping-based genome-size estimate of 1.3 Gb and inferred diploidy from a two-copy BUSCO peak. Both size figures are roughly twice the monoploid genome (Cx ≈ 540–650 Mb (Rejlová et al., 2019; Cronk et al., 2016; Hirabayashi et al., 2025)). We added this assembly as an exploratory target under a preregistration amendment written before it was downloaded. Only 17.1 % of RefSeq genes are single-copy in it; 51.0 % have two near-identical loci and 23.9 % have three or more, on contigs of median 40 Mb (0.06 % on contigs shorter than 1 Mb). The fair comparison is with a diploid assembled to both haplotypes: combining FH1 + FH2 or MH1 + MH2 gives ≥ 3 loci for 11.5 %–13.3 % of genes (gene families), and the two DToL tetraploid pseudo-haplotypes combined give 32.1 %. The Bonn value is about twice the unphased-diploid value and three-quarters of the tetraploid value, and its genes sit at diploid-level divergence from the Romanian female (nearest-copy R = 0.94 over 16,604 genes; f99 = 1.18 %; single-copy genes R = 1.11, CI 1.03–1.18), with Bonn-to-Kew divergence at the allelic level (median 0.59 %). The copy structure is intermediate between a pooled diploid and a pooled tetraploid and is what a tetraploid of the same lineage collapsed to about two copies per locus would produce; a nanopore assembly of a heterozygous diploid with some triplicated haplotigs could also produce it. The depositors' own diagnostics do not discriminate: 80.8 % duplicated BUSCOs with two copies dominant is expected both for a diploid assembled to both haplotypes and for a tetraploid collapsed to two; read-depth peaks at ≈ 25× and ≈ 50× (mean ≈ 40×) from 53.5 Gb of R9 reads mapped to the 1.1-Gb assembly imply either a diploid whose monoploid genome is ≈ 0.9 Gb or a tetraploid whose monoploid genome is ≈ 0.45 Gb — neither matches the flow-cytometric 0.55–0.62 Gb (Rejlová et al., 2019; Hirabayashi et al., 2025), so the depth calculation cannot be used as written; and the 1.3-Gb mapping-based genome size is doubled by the presence of both haplophases in the reference whatever the ploidy. The plant was grown from a single root, excluding pooling. Flow cytometry of plant XX-0-BONN-49798, or a k-mer spectrum of its reads, would settle the question (VALIDATION.md); we therefore report the Bonn plant's ploidy as unresolved:

tetraploidy is a possible, not an established, explanation.

Nucleotide diversity is depressed around every predicted centromere, most deeply on "polycentric" chromosomes (negative result)

The preregistered hypothesis H3 predicted a pericentromeric diversity trough on the five chromosomes predicted to be acrocentric (8, 9, 11, 12, 13) but not on the seven predicted to be polycentric (1–7). Using the mean of the three diploid pairwise distances per gene as $\hat{\pi}$ and the RepeatOBserver intervals of Hirabayashi et al. (2025) (Table S8) as centromere positions, all five acrocentric chromosomes showed the predicted gradient (Spearman $\rho > 0$ with circular-shift $p < 0.05$ in 5/5; mean trough ratio $T = 0.67$). But the polycentric chromosomes showed a deeper trough (mean $T = 0.38$; chromosome 10, "metacentric", $T = 0.40$), and the exact permutation over all 792 class labelings gave $p = 1.00$ in the predicted direction and $p = 0.001$ in the opposite direction — a significant effect opposite to the hypothesis (Figure 3). The two preregistered controls do not remove it: diversity falls with distance from the chromosome ends on polycentric chromosomes (mean $\rho = -0.37$) much more than on acrocentric ones (-0.11), i.e. the trough is central rather than centromere-anchored; and after removing all genes within 2 Mb of the predicted centromere point, $T = 0.74$ (acrocentric) vs 0.41 (polycentric). The held-out male replicate ($\hat{\pi} = d(MH1, MH2)$, male centromere predictions) gave $T = 0.77$ vs 0.44, reverse $p = 0.018$, and 0.75 vs 0.48 after 2-Mb exclusion. H3 is falsified: every nettle chromosome carries a broad low-diversity core — 10–25 Mb wide on the large chromosomes predicted to be polycentric — and the within-individual diversity landscape does not show the distributed recombination that true polycentricity would allow. We note that the male chromosome 13 centromere interval lies at the opposite end from the female one and coincides with locally high diversity, and that the 1,250,000-bp interval start recurring in both published tables is probably an analysis-window edge; the chromosome 13 call should be treated as unresolved.

The chr8/chr13 – *U. urens* chr2 relationship cannot be polarised (inconclusive result)

Reciprocal-best-hit synteny confirmed one-to-one correspondence between every *U. dioica* chromosome and a single *U. urens* chromosome except chromosomes 8 and 13, which both map to *U. urens* chromosome 2 (1,252 and 739 collinear-block genes), reproducing Hirabayashi et al. (2025). The preregistration named *Parietaria judaica*, ramie (*Boehmeria nivea*), *Morus notabilis* and *Cannabis sativa* as outgroups; ramie and *Morus* were dropped because no chromosome-level assembly of either carries an NCBI annotation (the *Morus notabilis* RefSeq annotation is on a scaffold-level assembly), a deviation we declare. Under the preregistered rule — direction called only if ≥ 70 % of the chr8-derived and of the chr13-derived block genes lie on one shared outgroup chromosome (fission in *U. dioica*) or on different chromosomes (fusion in *U. urens*) in two outgroups including one Urticaceae — neither direction was supported: in *Parietaria* the chr8 genes split between PJ1 and PJ3 (453/423) and the chr13 genes between PJ5 and PJ6 (263/219); in *Cannabis* chr8 genes fall on CAN1/CAN4/CAN8 and chr13 genes on CAN10/CAN3 (Figure 4b, c). The negative-control pair (chromosomes 4 and 6) mapped to disjoint outgroup chromosomes as required. Three observations qualify the result. *Parietaria* alternates between chromosome pairs at fine scale and has 35,329 annotated genes against 18,378 in *U. urens*, the signature of a lineage-specific genome duplication that splits one-to-one orthologs across two chromosomes and makes the ≥ 70 % rule unattainable, so "inconclusive" partly reflects outgroup choice. *Cannabis* is at Rosales distance and shows deep rearrangement. And *U. urens* chromosome 2 is not a simple end-to-end fusion: it is an eight-segment mosaic of chr8- and chr13-derived blocks, one inverted (Figure 4a), with no interstitial telomeric repeat arrays (all five (TTTAGGG)n≥4 arrays lie within 11 kb of one end), so at least seven breakpoints separate the two karyotypes on this chromosome and a single fission-versus-fusion polarisation is not a well-posed question. In both outgroups the chr8- and chr13-derived gene sets occupy entirely disjoint chromosomes, which under parsimony leans toward fusion plus inversions in the *U. urens* lineage, but this is offered as a lean, not a result.

Discussion

The question of whether the common tetraploid nettle is an auto- or an allopolyploid has stood since the mid-twentieth century. Within the sequence that has been deposited for the Kew tetraploid, the answer is clear: neither pseudo-haplotype contains a class of genes more divergent from the Eurasian

diploid than one Eurasian diploid is from another, whether those genes are examined as best loci, as retained second copies, as pairs between the tetraploid's own pseudo-haplotypes, or as spatial blocks along the chromosomes. The power model makes the statement quantitative: a second subgenome diverged by ≥ 2 % and present in ≥ 10 % of the deposited genes, or by ≥ 0.5 % in ≥ 25 %, is excluded. The deposited tetraploid is therefore drawn from the Eurasian diploid gene pool — an autotetraploid, or, indistinguishably by this assay, a tetraploid formed between diploid populations no more differentiated than present-day Romanian and Bulgarian nettles. This is consistent with the Hyb-Seq finding that tetraploids and diploids are not separable (Rejlová et al., 2021), extends the autopolyploid origin those authors proposed for Po-basin tetraploids to a north-west European tetraploid whose history they left unresolved, and agrees with the uniform monoploid genome size across the complex (Rejlová et al., 2019) and with the documented inter-cytotype gene flow through triploids (Rejlová et al., 2019); to our knowledge it is the first genome-wide, assembly-based measurement of subgenome divergence in the species, and the first to put a bound on the divergence an undetected second lineage could have.

Three features of the design give the conclusion its weight. First, the baseline is internal: the three diploid haplotypes come from two populations ~400 km apart, so b already includes whatever population structure exists among Eurasian diploids, and t is compared with b using the identical pair of reference haplotypes and the identical minimum-of-two statistic. Second, the test was preregistered with its thresholds, its held-out replicate, its controls and its artefact checks; the departures from the preregistration are listed in the Methods and none touches the decision statistics. Third, the result is replicated across the held-out alternate pseudo-haplotype, is insensitive to alignment trimming and to the paralog screen, and the Bonn assembly, whichever its ploidy, shows the same diploid-level divergence. An independent reproduction of the headline numbers from the archived BLAST tables, and from scratch on a random subset of genes, agreed with ours to the third decimal.

Two findings that emerged from review deserve comment. The small but real excess of t over b ($R \approx 1.10$, chromosome-level CI excluding 1) is expected under autopolyploidy with geography — the UK tetraploid is further from Romania than Bulgaria is, tetrasomic populations hold more allelic diversity, and the pseudo-haplotype's mixing of four haplotypes can only inflate best-locus divergence — but it is also exactly what a low-φ, low-δ contribution would produce, which is why we report the excluded region rather than a point verdict. And the spatial clustering of high-t genes, which the preregistration had treated as a signature of a shared subgenome, turns out to be produced by the baseline rather than by the tetraploid: over 20 Mb of chromosome 7 and 3 Mb of chromosome 8 the Romanian female, the Bulgarian male and the tetraploid alternate pseudo-haplotype share one nearly invariant haplotype, so that ordinary allelic divergence in the tetraploid primary becomes a 4-standard-deviation outlier against b. These blocks are themselves of interest — a shared Romanian–Bulgarian haplotype with $W \approx 0.15$ % across 20 Mb of chromosome 7 suggests either a recent sweep or an inversion polymorphism in the diploid, and the chromosome 8 block lies in the inversion-rich region already noted in the diploid assemblies (Hirabayashi et al., 2025, 2026) — but they say nothing about tetraploid origin. The lesson for assembly-based polyploid tests generally is that a divergence outlier must be decomposed into its numerator and denominator before it is read as evidence of a second lineage.

The reinterpretation of the Bonn assembly illustrates the cytotype trap that this species sets. A long-read primary assembly of a tetraploid with ~0.5 % allelic divergence collapses to about two copies at most loci, giving a two-copy BUSCO peak that looks diploid while the assembly span and mapping-based genome size are twice the monoploid value. Our copy-structure statistic — the fraction of RefSeq genes with ≥ 3 near-identical loci, compared with the same statistic for a haplotype-resolved diploid with both haplotypes pooled — offers a cheap first check for any new nettle genome; read-depth distributions at the triplicated genes or flow cytometry of the sequenced plant would settle it. Because 87 % of European nettles are tetraploid, future resequencing projects should expect tetraploids by default.

The negative result on centromere classes is informative in its own right. Nucleotide diversity in nettle coding sequences falls to near zero over a 10–25 Mb core on each of the seven large chromosomes predicted to be polycentric, and over narrower regions near the predicted centromeres of the acrocentric chromosomes. The within-individual diversity landscape, which integrates recombination and linked selection over the coalescent history of the population, therefore looks like that of a conventional monocentric karyotype with large pericentromeric low-recombination domains — or like a holocentric with strongly distal crossovers (Castellani et al., 2024) — and does not reveal the

distributed centromere organisation that RepeatOBserver repeat patterns suggest. Three readings are possible: the diffuse repeat patterns reflect large pericentromeric heterochromatin blocks rather than multiple kinetochore sites, as in legume metapolycentromeres where genes and repeats are uniformly distributed across an 80-Mb centromere (Macas et al., 2023); the chromosomes are truly polycentric but crossovers are nevertheless confined to distal euchromatin, as the two-event crossover count on polycentric chromosomes in the parent–offspring pair would suggest (Hirabayashi et al., 2026); or the diversity trough is driven by gene density and background selection independently of centromere type. The analysis cannot separate these, and CENH3 chromatin immunoprecipitation or immunostaining remains the decisive experiment (VALIDATION.md). What the data do exclude is a nettle karyotype in which the polycentric chromosomes recombine uniformly along their length.

The chr8/chr13 analysis shows why chromosome identity is a poor homology unit across *Urtica* (Hirabayashi et al., 2025): the *U. urens* homologue is an eight-segment mosaic, so at least a fusion/fission and several inversions separate the two species on this chromosome alone, and the available outgroups — one apparently polyploid, one at Rosales distance — do not retain the ancestral arrangement. A chromosome-level, annotated *Boehmeria* or *Laportea* genome would be the right outgroup.

Limitations

The principal limitation is that the test compares deposited assemblies, not reads. The DToL primary and alternate pseudo-haplotypes together contain 68 % of the tetraploid's reliable read k-mers (Christenhusz et al., 2025); the remaining third was removed during purging and curation. If the tetraploid were an allopolyploid whose second subgenome was diverged enough to be separated cleanly by the assembler and then purged as haplotigs, that subgenome would be absent from every tier we examined — best loci, second loci and the alternate pseudo-haplotype — and our test would return exactly the result we observe. The k-mer completeness figures themselves do not settle the matter: under a simple neutral model (one of our reviewers' calculation, $k = 31$, $\pi = 0.5$ %) two haplotypes of an autotetraploid would be expected to hold ≈ 88 % of read k-mers, and two haplotypes of an allotetraploid with a second pair diverged by 1–1.5 % ≈ 64–75 %, so the observed 68 % is closer to the latter; but the same model ignores collapsed repeats (69 % of the genome is transposable-element sequence), the lower per-haplotype coverage of the tetraploid (44.9 Gb of HiFi, "42-fold" by the note's own reckoning, i.e. ~19× per haplotype), and the 88 haplotypic duplications removed by curation, each of which lowers completeness without any second lineage. We consider a hidden divergent subgenome unlikely: with ~10 % species-level divergence a diverged subgenome would assemble as separate contigs that purge_dups would treat as haplotigs only where they are collinear and ≥ 80–90 % identical, retained second loci show no excess at 90–97 % identity, and the Bonn assembly, which was not purged, shows the same divergence profile; but it cannot be excluded from assemblies alone, and the decisive test is read-based — a k-mer spectrum (GenomeScope2/Smudgeplot) of the DToL HiFi reads, which under autotetraploidy shows AAAB and AABB smudges with tetraploid-level heterozygosity and under allotetraploidy a diploid-like AB peak at the subgenome divergence, plus a per-gene read-depth profile mapped to the diploid. We specify this in VALIDATION.md; it requires about 60 Gb of reads and did not fit the resource ceiling of this study. All statements of exclusion in this paper are therefore conditional on the deposited sequence.

Further limitations: the test is blind to allopolyploidy between lineages less differentiated than present-day Romanian and Bulgarian diploids, and to homogenisation by gene conversion, which could erase subgenome identity only over timescales far longer than the retained copy structure suggests. The diploid baseline derives from one female and one paternal haplotype; a population sample of diploids would sharpen Q99(b), and the shared low-diversity blocks on chromosomes 7 and 8 show that a two-population baseline can be locally deflated. The DToL plant's cytotype rests on the genome note's statement (the note does not report flow cytometry; the plant's sex, male, is recorded in BioSample SAMEA7521824), and the Bonn plant's on our copy-structure inference. The preregistered dip test could not be evaluated as written (the raw-scale test is non-diagnostic for zero-inflated discrete distances) and is reported on the log scale as a post-hoc clarification. Centromere positions are RepeatOBserver predictions without cytological validation, and the chromosome 13 call is unresolved. All conclusions are restricted to coding sequences of single-copy genes.

Methods

Data. Assemblies and annotations were downloaded from the NCBI genomes FTP on 2026-10-23 with MD5 verification (Table 1): *U. dioica* female haplotypes GCA_048565285.1 (FH1) and GCA_048565295.1 (FH2) (Hirabayashi et al., 2025); male haplotypes GCA_054948195.1 (MH1) and GCA_054948205.1 (MH2) with RefSeq annotation GCF_054948205.1 release RS_2026_09 (released 2026-10-20) (Hirabayashi et al., 2026); DToL drUrtDioi1.1 primary GCA_964188135.1 (DP) and alternate GCA_964188125.1 (DA), BioSample SAMEA7521824 (male, RBG Kew, collected 2020-08-05), assembled with HiCanu, purge_dups and YaHS (Christenhusz et al., 2025); *U. urens* drUrtUren1.1 GCA_958296335.1 with Ensembl Rapid Release geneset 2024_02 (Christenhusz et al., 2024b); *Parietaria judaica* drParJuda1.1 GCA_954871525.1 with Ensembl geneset 2024_02 (Christenhusz et al., 2024a); *Cannabis sativa* cs10 GCF_900626175.1 RefSeq proteins and GFF. The Bonn assembly (Urtica_dioica.v1, bonndata doi:10.60507/FK2/1XHSZ3, CC BY 4.0) (Wolff et al., 2026) was downloaded with MD5 verification. Parental origin of male chromosomes follows Table 2 of Hirabayashi et al. (2026) (maternal = MH1 for chromosomes 1, 2, 4, 7–13 and MH2 for 3, 5, 6). Putative centromere intervals were taken from Table S8 of Hirabayashi et al. (2025) (female) and Supplementary Table 5 of Hirabayashi et al. (2026) (male); intervals < 100 kb were ignored, the centromere point of an acrocentric chromosome was the midpoint of its largest interval and that of a polycentric or metacentric chromosome the midpoint of the span of its intervals; this rule was fixed before any divergence value was computed.

Preregistration. Hypotheses, statistics, thresholds, controls, held-out data and artefact checks were written to notes/preregistration.md (SHA-256 recorded) at 2026-10-23T08:34:58+01:00, before any FASTA was downloaded; one amendment adding the Bonn assembly as an exploratory target was written before that file was downloaded. Both versions are supplied.

Query set. From GCF_054948205.1_cds_from_genomic.fna.gz we kept, per protein-coding gene, the longest CDS with length ≥ 300 bp, length divisible by 3 and no ambiguous base (23,131 queries, 23,030 on chromosomes).

Homology search. Each genome was decompressed and indexed with makeblastdb (BLAST+ 2.17.0, Windows x64 build (Camacho et al., 2009)). Queries were searched with blastn -task dc-megablast, evalue 1e-20, max_target_seqs 5, max_hsps 40, dust off, soft masking off, 8 threads, outfmt 6 with qseq/sseq. HSPs were grouped into loci by subject sequence and strand, splitting at gaps > 50 kb; within a locus, HSPs were accepted greedily by bitscore if they overlapped ≤ 20 % with already-accepted query intervals; locus score = summed bitscore; coverage = fraction of CDS positions covered. A gene was single-copy in a target if the second-best locus scored < 0.8 × the best.

Query-anchored alignments and distances. For the best locus, the target base aligned to each CDS position was recorded (deletions as gaps; target insertions dropped), discarding the first and last 10 aligned positions of every HSP that did not reach the CDS end (trim calibrated on the MH2 self-search, which reduced the self-distance floor from 0.00079 to 0.00026 while retaining 93 % coverage), and masking positions where the self-search did not recover the query base. The p-distance between two targets is the number of mismatches divided by the number of columns where both carry A/C/G/T (≥ 300 columns required). Distances were computed for all target pairs (code/h1_distances.py). A full re-run with no trimming (code/gene_divergence.py with NETTLE_TRIM=0) is reported as a sensitivity analysis.

H1 statistics (preregistered). Genes required coverage ≥ 0.8 and single-copy status in FH1, FH2, MH1, MH2 and the tetraploid target, and a chromosome assignment. b = min[d(PAT, FH1), d(PAT, FH2)]; t = min[d(T, FH1), d(T, FH2)]; R = median(t)/median(b); f99 = fraction of genes with t > Q99(b). Decision rule: same-lineage if 0.67 ≤ R ≤ 1.5 and f99 ≤ 0.05 and no second mode in d(DP, DA) above Q99(b); allotetraploid (shared subgenome) if f99 ≥ 0.20 with spatial clustering of high-t genes and bimodal d(DP, DA); allotetraploid (both subgenomes divergent) if R ≥ 2 and f99 ≥ 0.50; otherwise inconclusive. Held-out: the alternate pseudo-haplotype DA was not examined until the DP analysis was recorded. Controls: technical floor min[d(MAT, FH1), d(MAT, FH2)]; scale d(·, *U. urens*).

H1 statistics added or changed after independent review (code/h1_test_v2.py). (i) Confidence intervals: 1,000 bootstrap resamples of whole chromosomes (primary) and of 2-Mb blocks of genes along MH2 (seed 20261023). (ii) Paralog screen: genes whose tetraploid best locus is > 5 percentage

points more divergent from *U. urens* than the diploid ortholog were removed (6 in DP, 15 in DA; statistics are reported with and without the screen in the result JSON). (iii) Spatial clustering: the preregistered Moran's I against circular shifts was found to be degenerate for a rare indicator (the statistic is nearly shift-invariant) and was replaced by the count of adjacent high-t gene pairs along chromosomes against 2,000 within-chromosome permutations of gene labels. (iv) A windowed scan of the paired difference t − b (50-gene rolling median) against the 99th percentile of the genome-wide maximum over 300 within-chromosome permutations of gene order, with the medians of t, b, W and the fraction of b = 0 reported for each block. (v) Dip test: Hartigan's dip (Hartigan & Hartigan, 1985) (diptest 0.8.5) on raw and on log10-transformed non-zero distances for d(DP, DA), W and b. (vi) Power model: for each (δ, φ) on a grid, b was resampled with replacement to the observed n, δ added to a random fraction φ of genes, and R and f99 recomputed and averaged over 20 replicates; a combination is excluded if its expected R or f99 exceeds the observed chromosome-level upper 95 % limit. (vii) Symmetry check: R on chromosomes 3, 5 and 6 (PAT = MH1) versus the others.

Copy structure. For each query and target, loci with coverage ≥ 0.8 and mean HSP identity ≥ 97 % were counted (code/copy_structure.py); unphased-diploid baselines pool the loci of FH1 + FH2 and of MH1 + MH2, and the tetraploid baseline pools DP + DA. Contig lengths of third-plus loci in the Bonn assembly were taken from the assembly index.

H3 (preregistered). π̂ per gene = mean of d(FH1, FH2), d(PAT, FH1), d(PAT, FH2); positions on FH1. Per chromosome: Spearman ρ between π̂ and distance to the centromere point, with a 1,000-shift circular null; trough ratio T = median π̂ of the 20 % of genes nearest the point / median of the rest. Class comparison: difference in mean T between the five acrocentric and seven polycentric chromosomes (chromosome 10 reported separately) against the exact distribution over all 792 label assignments, reported in both directions. Decision rule: supported if ρ > 0 with $p < 0.05$ in ≥ 4/5 acrocentric chromosomes, mean T(acro) < 0.8 and permutation $p < 0.05$; falsified if ≤ 2 acrocentric gradients or permutation $p > 0.2$. Preregistered controls: ρ with distance to the nearest chromosome end, and T recomputed after excluding genes within 2 Mb of the centromere point (added to the code after review; the preregistration named both). Held-out: the male assembly with π̂ = d(MH1, MH2) and male centromere predictions.

H2 (preregistered). Longest protein per gene for *U. dioica* (RefSeq), *U. urens* and *P. judaica* (Ensembl) and *C. sativa* (RefSeq) (code/prep_proteomes.py); reciprocal best BLASTP hits (evalue 1e-10, ≥ 50 % mutual coverage); collinear blocks of ≥ 5 pairs within 20-gene windows on both genomes (code/h2_synteny.py). S8 and S13 = *U. dioica* chr8 and chr13 genes in blocks with *U. urens* chr2. Direction called if ≥ 70 % of S8 and of S13 block genes lie on a shared (fission) or on different (fusion) outgroup chromosomes in ≥ 2 outgroups including ≥ 1 Urticaceae; negative-control chromosome pair drawn with seed 20261023. Telomeric arrays (TTTAGGG)n or (CCCTAAA)n with n ≥ 4 were located by regular expression on *U. urens* chromosome 2. Deviation: ramie and *Morus notabilis*, named in the preregistration, were not used (no annotated chromosome-level assembly of either on NCBI).

Declared departures from the preregistration. dc-megablast rather than megablast for all targets; HSP end-trimming and self-masking (fixed before any between-haplotype distance was seen); log-scale dip test; Moran's I replaced by adjacent-pair permutation; paralog screen; chromosome-level bootstrap; windowed paired scan and power model (added after review); outgroups dropped from H2; Bonn assembly added by amendment. None changes the preregistered decision statistics R and f99 by more than 0.01 or 0.1 percentage point.

Independent review. Before finalisation the analysis was given to two independent agents with no access to this manuscript's conclusions: one reproduced R and f99 for DP and DA from the archived BLAST tables with its own code and from scratch (new BLAST searches) for 150 randomly chosen genes (per-gene Pearson r 0.996–0.998, median absolute difference 0); the other performed a hostile review. Their reports and our point-by-point responses are supplied (paper/REVIEWER_RESPONSES.md).

Software and reproducibility. Python 3.11 with numpy 2.4.6, pandas 2.3.3, scipy, statsmodels 0.15, diptest 0.8.5, Biopython 1.88, matplotlib; BLAST+ 2.17.0 (ncbi-blast-2.17.0+-x64-win64.tar.gz, MD5 verified); Windows 11 host, 4 cores / 8 threads, 16 GB RAM. Total wall time ≈ 30 h; peak disk ≈ 14 GB. The Snakemake workflow (code/Snakefile, code/environment.yml) regenerates every table and figure from the accessions above, processing one assembly at a time and deleting decompressed FASTA after indexing. No read data were downloaded or aligned.

Data availability

All input data are public: NCBI assemblies GCA_048565285.1, GCA_048565295.1, GCA_054948195.1, GCA_054948205.1 / GCF_054948205.1 (annotation RS_2026_09), GCA_964188135.1, GCA_964188125.1, GCA_958296335.1, GCA_954871525.1, GCF_900626175.1; Ensembl Rapid Release genesets 2024_02 for *U. urens* and *P. judaica*; figshare project 230981 (supplementary tables, CC BY 4.0); bonndata doi:10.60507/FK2/1XHSZ3 (CC BY 4.0); Europe PMC supplementary files for PMC11722821. Per-gene distance tables, synteny blocks, result JSON files, figures, the preregistration with hashes, the laboratory notebook, the claim ledger, the independent reviewers' reports and all code are provided as a supplementary archive and in the companion repository https://github.com/edward-mason/urtica-tetraploid-origin (archived on Zenodo at posting; DOI to be inserted). Raw BLAST tables (~5 GB) regenerate from the workflow.

Competing interests

None declared.

Tables

Table 1. Assemblies used.

code	accession	individual	cytotype (source)	span	annotation used
FH1 / FH2	GCA_048565285.1 / GCA_048565295.1	female clone 11-4, Rovinari, Romania	diploid, 2n = 26 (chromosome count, (Hirabayashi et al., 2025))	575 / 521 Mb	— (BRAKER3 sets exist; not used)
MH1 / MH2	GCA_054948195.1 / GCA_054948205.1	male clone 48M, son of 11-4 × Bulgarian male 7-5	diploid ((Hirabayashi et al., 2026))	567 / 534 Mb	RefSeq RS_2026_09 on MH2 (23,586 genes)
DP / DA	GCA_964188135.1 / GCA_964188125.1	male, Kew, UK (DToL); HiCanu + purge_dups + YaHS	tetraploid; "one subgenome of the tetraploid genome"; k-mer completeness 55 % / 49 % (68 % combined) ((Christenhusz et al., 2025))	618 Mb / fragmented	—
BN	bonndata 10.60507/FK2/1XHSZ3	plant XX-0-BONN-49798, collected in Braunschweig, Germany; grown from a single root in Bonn	stated diploid (authors read the 1.1 Gb as two haplophases); ploidy unresolved, tetraploidy possible (this study)	1.1 Gb (hifiasm, ONT R10)	—
UU	GCA_958296335.1	*U. urens*, DToL	diploid, n = 12	340 Mb	Ensembl 2024_02 (18,378 genes)
PJ	GCA_954871525.1	*Parietaria judaica*, DToL	—	—	Ensembl 2024_02 (35,329 genes)
CAN	GCF_900626175.1	*Cannabis sativa* cs10	diploid, n = 10	—	RefSeq (25,297 genes)

Table 2. H1 statistics (paralog-screened; chromosome-level bootstrap CIs).

target	n genes	median b	median t	R (95 % CI)	f99 (95 % CI)	f95	adjacent high-t pairs obs / exp (p)	verdict
DP (discovery)	13,432	0.449 %	0.496 %	1.106 (1.061–1.163)	1.62 % (1.36 %–2.06 %)	6.8 %	30 / 3.9 (0.0005)	same-lineage
DA (held-out)	13,404	0.455 %	0.499 %	1.097 (1.065–1.137)	1.79 % (1.45 %–2.20 %)	6.7 %	46 / 4.5 (0.0005)	same-lineage
DP, no HSP trimming (sensitivity)	13,539	0.443 %	0.494 %	1.114	1.74 %	—	—	same-lineage

target	n genes	median b	median t	R (95 % CI)	f99 (95 % CI)	f95	adjacent high-t pairs obs / exp (p)	verdict
BN (exploratory, single-copy genes)	2,706	—	—	1.108 (1.027–1.185)	1.81 %	—	—	(same-lineage)
BN (exploratory, nearest copy, all genes)	16,604	—	—	0.940	1.18 %	—	—	(same-lineage)

Table 3. Power model: expected R and f99 for a second subgenome contributing a fraction φ of genes with extra divergence δ, against observed upper 95 % limits (R ≤ 1.163, f99 ≤ 2.06 %).

δ \ φ	0.1	0.25	0.5	0.75	1
0.25 %	R 1.06, f99 1.1 % (not excluded)	R 1.15, f99 1.1 % (not excluded)	R 1.30, f99 1.2 % (**excluded**)	R 1.43, f99 1.2 % (**excluded**)	R 1.56, f99 1.3 % (**excluded**)
0.50 %	R 1.11, f99 1.1 % (not excluded)	R 1.32, f99 1.2 % (**excluded**)	R 1.62, f99 1.3 % (**excluded**)	R 1.89, f99 1.5 % (**excluded**)	R 2.12, f99 1.7 % (**excluded**)
1.00 %	R 1.14, f99 1.3 % (not excluded)	R 1.46, f99 1.6 % (**excluded**)	R 2.38, f99 2.3 % (**excluded**)	R 2.89, f99 2.9 % (**excluded**)	R 3.23, f99 3.5 % (**excluded**)
1.50 %	R 1.14, f99 1.9 % (not excluded)	R 1.46, f99 3.3 % (**excluded**)	R 3.34, f99 5.6 % (**excluded**)	R 3.97, f99 7.9 % (**excluded**)	R 4.34, f99 10.3 % (**excluded**)
2.00 %	R 1.14, f99 4.0 % (**excluded**)	R 1.45, f99 8.3 % (**excluded**)	R 4.46, f99 15.9 % (**excluded**)	R 5.07, f99 23.5 % (**excluded**)	R 5.46, f99 30.9 % (**excluded**)
3.00 %	R 1.14, f99 11.0 % (**excluded**)	R 1.45, f99 25.6 % (**excluded**)	R 6.29, f99 50.4 % (**excluded**)	R 7.29, f99 75.2 % (**excluded**)	R 7.68, f99 100.0 % (**excluded**)
5.00 %	R 1.14, f99 10.9 % (**excluded**)	R 1.46, f99 25.8 % (**excluded**)	R 9.53, f99 50.5 % (**excluded**)	R 11.75, f99 75.3 % (**excluded**)	R 12.14, f99 100.0 % (**excluded**)

Figure legends

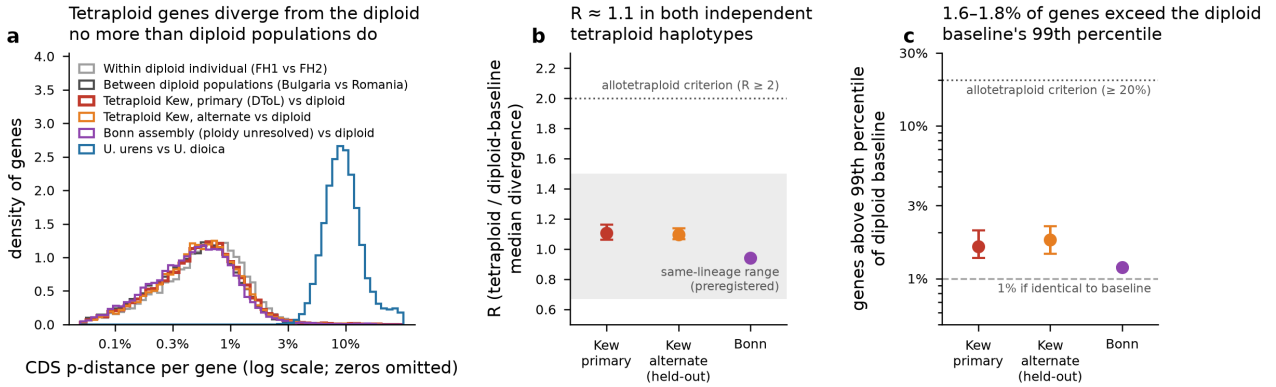


Figure 1. The tetraploid pseudo-haplotype lies inside the diploid gene pool. (a) Density of per-gene CDS p-distances (log scale, zeros omitted) for: the two haplotypes of the Romanian diploid female (W, n = 16,819); the Bulgarian diploid paternal haplotype vs its nearest female haplotype (b); the DToL tetraploid primary (t, n = 13,432) and alternate (n = 13,404) pseudo-haplotypes vs their nearest female haplotype; the Bonn assembly's nearest copy vs the female (n = 16,604); and *U. urens* vs *U. dioica* (scale). (b) R = median t / median b with chromosome-level bootstrap 95 % CIs; grey band, preregistered same-lineage range; dotted line, allotetraploid criterion. (c) Fraction of genes with t above the 99th percentile of b, log scale, with CIs; dashed line, 1 % expected under identity; dotted line, allotetraploid criterion. Bonn values are exploratory.

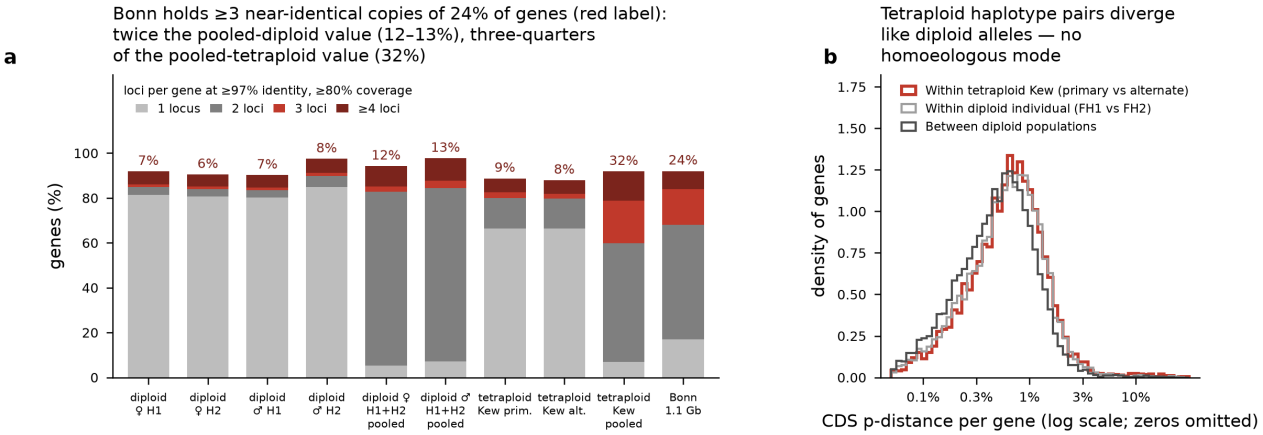


Figure 2. Copy structure and within-tetraploid divergence. (a) Per assembly or pooled pair of assemblies, percentage of RefSeq genes with 1, 2, 3 or ≥ 4 loci at ≥ 97 % identity and ≥ 80 % CDS coverage; the label gives the percentage with ≥ 3 loci. (b) Density of per-gene distances between the DToL primary and alternate pseudo-haplotypes (n = 13,246) compared with W and b.

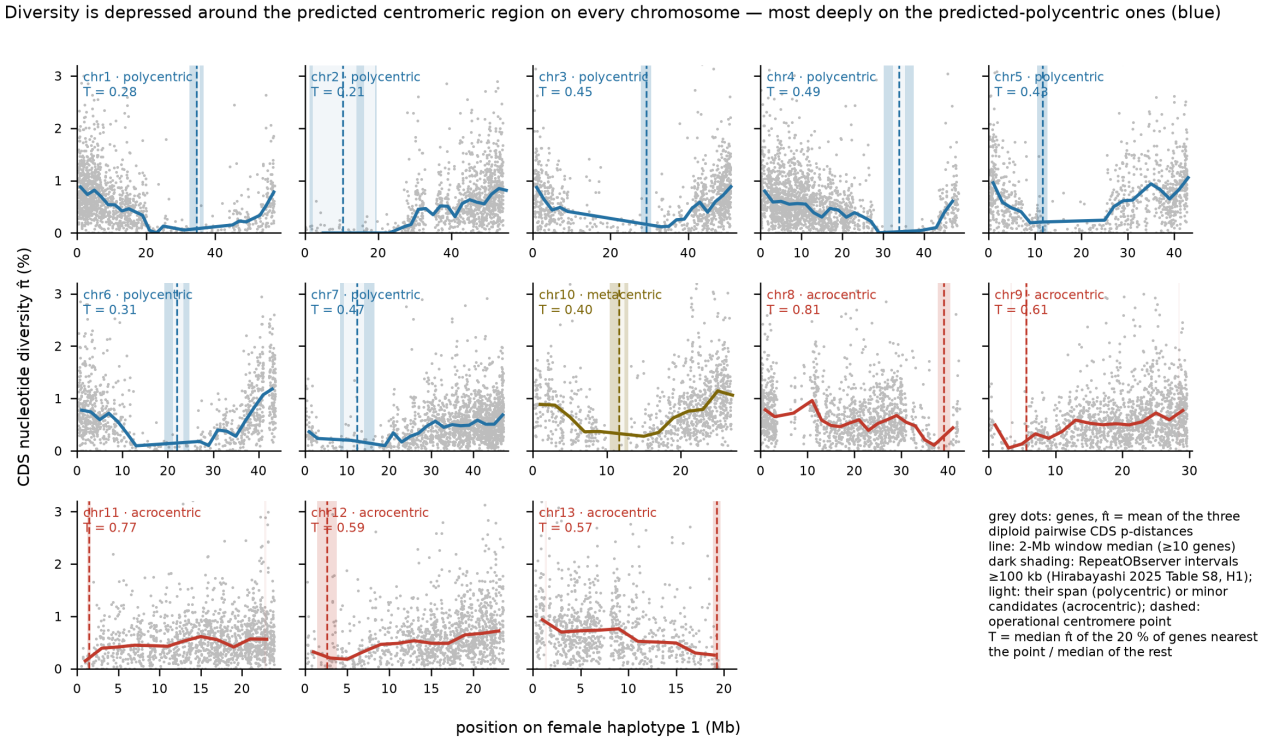


Figure 3. Nucleotide diversity along the 13 female chromosomes. Grey points: π̂ per gene (mean of the three diploid pairwise CDS p-distances); line: 2-Mb window median (≥ 10 genes); dark shading: RepeatOBserver centromeric intervals ≥ 100 kb (Hirabayashi et al. 2025, Table S8, haplotype 1); light shading: their span (polycentric) or minor candidate intervals (acrocentric); dashed line: operational centromere point. T = median π̂ of the 20 % of genes nearest the point divided by the median of the rest. Blue, predicted polycentric; red, predicted acrocentric; olive, chromosome 10.

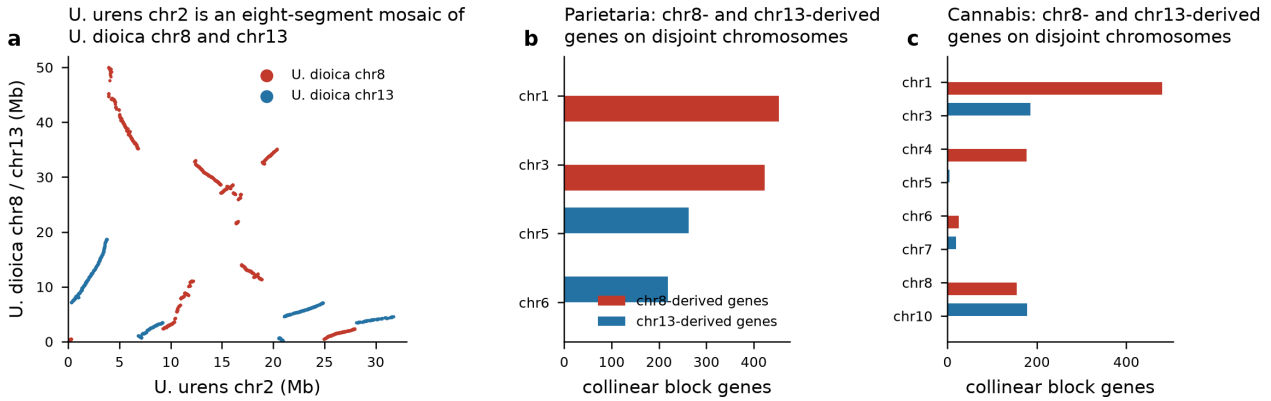


Figure 4. Synteny of *U. dioica* chromosomes 8 and 13. (a) Positions of collinear-block gene pairs between *U. urens* chromosome 2 and *U. dioica* chromosome 8 (red) or 13 (blue). (b, c) Number of chr8-derived and chr13-derived block genes on each *Parietaria judaica* and *Cannabis sativa* chromosome (chromosomes carrying ≥ 5 block genes shown).

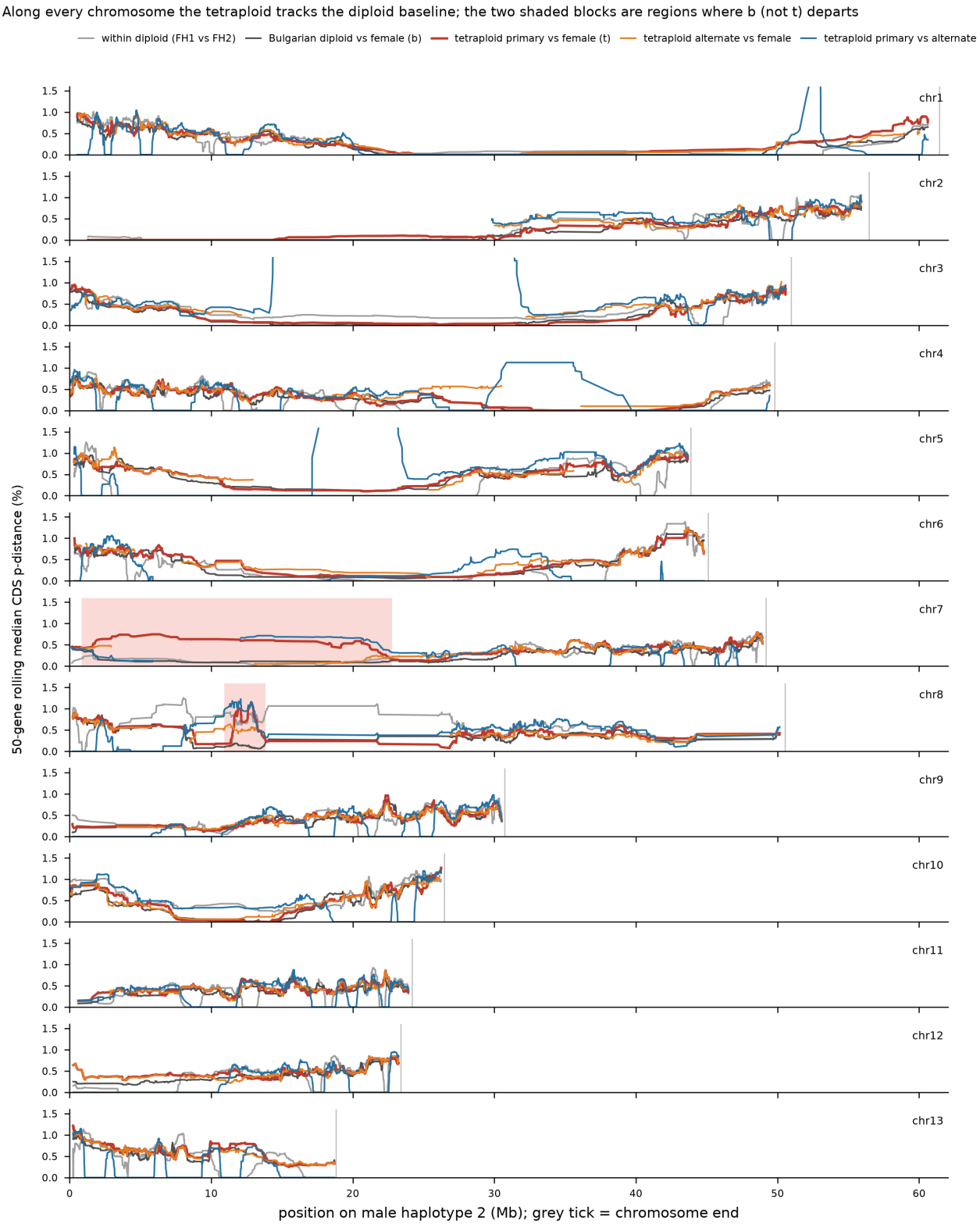


Figure 5. Windowed divergence along the 13 chromosomes. 50-gene rolling medians (positions on male haplotype 2) of W (FH1 vs FH2), b (Bulgarian paternal haplotype vs nearest female haplotype), t (tetraploid primary vs nearest female haplotype), the same for the tetraploid alternate, and primary vs alternate. Shaded: the two blocks in which the rolling median of t − b exceeds the within-chromosome permutation 99th percentile; in both, b is deflated and t is at the genome-wide diploid level.

AI disclosure statement

Tools. Claude Science (Anthropic), an agentic research environment powered by the large language model Claude (model identifier claude-fable-5-1), used between 2026-10-22 and 2026-10-24 on the author's Windows 11 workstation. Sub-agents of the same system performed the literature inventory, the public-data inventory, an independent reproduction (Reviewer A) and a hostile review (Reviewer B) of the result.

What the AI produced. The AI generated the research plan, the hypothesis list and scoring, the preregistration text (including thresholds), all analysis code (Python, BLAST+ invocations, Snakemake workflow), all figures, the laboratory notebook, the literature map, the claim ledger, the validation plan and the full first draft of this manuscript, including the abstract, results, discussion, methods and reference list. Numerical results were produced by code written and executed by the AI on public data; reference metadata were retrieved programmatically from CrossRef and every DOI was machine-verified to resolve.

What the human author did. The human author defined the mission, constraints and quality criteria, approved the plan, granted data-access permissions, and is responsible for reading, checking and, where necessary, correcting every statement before submission. The human author is the sole author of record and takes full responsibility for the content; generative AI is not an author and does not meet authorship criteria.

Known limitations of AI involvement. Literature coverage depends on programmatic database queries (OpenAlex, CrossRef, Europe PMC, Semantic Scholar, bioRxiv) and may miss non-indexed theses or grey literature; the sub-agent literature report explicitly declared that a thesis search was not performed and that several older sources were characterised from abstracts only. The AI's interpretation of published tables (e.g., centromere interval tables) was mechanical and is documented in the notebook. Text passages were written by the AI from the numerical results and should be read as such.

Independent review inside the system. Two sub-agents with fresh contexts, no access to the draft manuscript and separate working directories reviewed the work (paper/reviews/): Reviewer A reproduced the headline statistics from archived tables and from new BLAST searches; Reviewer B performed a hostile review and its own recomputations. Their reports are supplied verbatim and every point and the action taken is listed in paper/REVIEWER_RESPONSES.md. The mission brief also asked for comparative-plant-genomics criteria to be added to the platform's Reviewer specialist (Settings → Specialists → Reviewer); that is a user-interface setting the AI cannot change and was not done — the reviewer agents were instead given explicit domain criteria in their task text.

What the AI did not do. It did not download or analyse raw sequencing reads, run flow cytometry, or access any non-public data; it did not read the full text of Rejlová et al. 2021, Geltman 1990 or the Rejlová PhD thesis (closed access / not online), which are cited from abstracts or through secondary sources, as stated in the text. All web-search results used for the novelty check are listed in notes/notebook.md (2026-10-24).

Human verification required before submission. Author name and affiliation; confirmation that every cited quotation matches the source; a read of the full Rejlová et al. 2021 paper for any auto/allo analysis; optional execution of VALIDATION.md item 5 (k-mer analysis of the DToL reads), which would materially strengthen or overturn Claim 1.

Supplementary material

Supplementary files accompany the preprint as a single archive (see Data availability): S1 preregistration (v1 and v2 with SHA-256 hashes); S2 laboratory notebook; S3 independent reviewer reports and point-by-point responses; S4 claim ledger; S5 validation plan; Table S1 per-gene distances (gene_distances.csv); Table S2 per-gene H1 statistics (h1v2_genes_DP.csv, h1v2_genes_DA.csv); Table S3 per-chromosome H3 statistics; Table S4 synteny blocks; Table S5 copy-structure and sensitivity tables (JSON); code (Snakemake workflow, Python scripts, environment.yml).

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